SI 315

Assignment 6

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Due at 11:59 pm on Wednesday, November 4

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1 Problem 1: Auctions

1.1) Expected Revenue

Problem. In this problem we will ask how the *number of bidders* in a second-price, sealed-bid auction affects how much the seller can expect to receive for his object. Assume that there are *two bidders* who have independent, private values v_i which are either 1 or 3. For each bidder, the probabilities of 1 and 3 are both 1/2. (If there is a tie at a bid of x for the highest bid, the winner is *selected at random* from among the highest bidders and the price is x).

1.1.I) Expected Revenue with 2 Bidders

Problem. Show that the seller's expected revenue is 6/4.

Solution. We examine the four equally-likely possible cases for (v_0, v_1): (1, 1), (3, 1), (1, 3), and (3, 3). In all but the last, (3, 3), the final price is 1, with the price of the last case being 3. So, the expected payout for the seller is $\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 3 = \frac{6}{4}$.

1.1.II) Expected Revenue with 3 Bidders

Problem. Now let's suppose that there are three bidders who have independent, private values v_i which are either 1 or 3. For each bidder, the probabilities of 1 and 3 are both 1/2. What is the seller's expected revenue in this case?

Solution. There are eight cases:

B1	B2	B3	Seller's revenue
1	1	1	1
1	1	3	1
1	3	1	1
1	3	3	3
3	1	1	1
3	1	3	3
3	3	1	3
3	3	3	3

So the expected revenue is $(4 \cdot 1 + 4 \cdot 3)/8 = 2$.

1.1.III) Explanation

Problem. Briefly explain why changing the number of bidders affects the seller's expected revenue.

Solution. There are only two possible selling prices: 1 and 3, with 3 only being possible when there are at least two bidders both bidding 3. This is unlikely with only two bidders ($p = 0.25$). But, increasing the number of bidders increased the likelihood of having at least two bidders with a value of 3 ($p = 0.5$), which increased the expected value of the second-highest bid.

1.2) Inventing Auctions

Problem. Give an example of an auction (i.e., a set of bidders, true values, and bids) with 5 bidders that satisfies each property. If there is a tie at a bid of x for the highest bid the winner is selected at random from among the highest bidders and the price is x. Explain why your auction satisfies the desired property.

Note: Each property is independent (so you need a different example for each part, and the example in, say, part (a) only needs to satisfy property (a)).

1.2.I) First-Price Same as Second-Price

Problem. All bidders have the same payoff from a first-price sealed-bid auction as from a second-price sealed-bid auction.

Solution. This happens when the first price is the same value as the second price. In this case, all bidders who lose the bid get the same payoff of zero and the winner still pays the same price (assuming the same bidder wins by chance in both cases).

For example:

Bidder #	Value	Bid	Payoff
1	6	**4**	2
2	5	**4**	0
3	4	3	0
4	4	3	0
5	3	2	0

Either way, the winner (bidder 1) pays the same final price (and, therefore, has the same payoff) because the first and second bids are the same.

1.2.II) All-Pay Auction

Problem. The average payoff in a sealed-bid all-pay auction (i.e., whoever bids the highest wins the item, but everyone has to pay their own bid) is 5.

Solution. This can only happen where the winner bids significantly below their true value, enough to account for the losses of all other bidders. The total payoff of the entire bidding pool has to be 25 to make an average payoff of 5 per bidder. For example, let the winner have a payoff of 45 and let all other bidders have a payoff of -5. This would sum to $45 - 4 \cdot 5 = 25$ and make an average payoff of $25/5 = 5$.

We can make this happen with the following auction:

Bidder #	Value	Bid	Payoff
1	100	55	45
2	90	5	-5
3	80	5	-5
4	70	5	-5
5	60	5	-5

It doesn't really matter the values of the losing bidders, they only need to all lose and pay a total of $20.

2 Problem 2: Third-Price Sealed-Bid Auction

Problem. In a third-price sealed-bid auction, where the buyer with the highest bid buys the item and pays the third-highest bid, is bidding your true value always a dominant strategy? If yes, provide a proof. If not, provide a counterexample.

3 Problem 3: Procurement Auctions

Problem. A city wants to hire one contractor. Each contractor submits a sealed bid. Each contractor also has a publicly known quality credit. The quality credit represents the monetary value the city assigns to the contractor's quality, such as expected savings from fewer future repairs or lower maintenance costs.

To account for differences in quality, the city calculates each contractor's adjusted bid as follows:

$$\text{Adjusted bid} = \text{Bid} - \text{Quality credit}$$

The city awards the contract to the contractor with the lowest adjusted bid. The winning contractor's payoff is the payment received minus their true cost, while a losing contractor's payoff is 0. Assume bids must be whole dollars.

Contractor	True cost	Quality credit
A	\$70	\$30
B	\$58	\$15
C	\$49	\$5
D	\$45	\$0

a. Suppose all contractors bid their true costs. Compute each contractor's adjusted bid and determine who wins. Does the contractor with the lowest submitted bid win?

b. Now consider the procurement version of a first-price sealed-bid auction: the contractor with the lowest adjusted bid wins and is paid their own bid (not the adjusted bid).

   Suppose B, C, and D bid their true costs, and A knows this. Holding the other contractors' bids fixed, what bid should A submit to maximize A's payoff?

   Next, suppose instead that A, C, and D bid their true costs, and B knows this. Would B want to submit a bid low enough to beat A's adjusted bid? Explain your answer in terms of B's payoff. Explain.

c. Now consider the procurement analog of a second-price sealed-bid auction. The contractor with the lowest adjusted bid wins, but instead of receiving their own submitted bid, the winner is paid based on the second-lowest adjusted bid and their own quality credit.

   Specifically, the winner's payment is calculated as:

   $$\text{Winner's payment} = \text{Second-lowest adjusted bid} + \text{Winner's quality credit}$$

   For example, suppose the lowest adjusted bid is \$40, the second-lowest adjusted bid is \$50, and the bidder with the lowest adjusted bid has a quality credit of \$20. The winner (the lowest adjusted bidder) would receive \$50 (the lowest adjusted bid) + \$20 (the winner's quality credit) = \$70.

   Suppose everyone bids their true cost, but B's quality credit is now x, where $0 \leq x \leq 25$. All other quality check values are as shown in the table. Determine who wins and how much the winner is paid as a function of x. What happens at the value of x where A and B tie?

4 Problem 4: Market-Clearing Prices

Problem. The tables below show the valuation of 3 buyers x, y, z for 3 different houses sold by A, B, C. For each table, find the market-clearing prices following the procedure presented in class. Show the price of each house and the preferred-seller graph at the end of each round of the procedure.

a.

Buyer	Valuation for A's house	Valuation for B's house	Valuation for C's house
x	2	2	1
y	1	5	3
z	2	3	0

b.

Buyer	Valuation for A's house	Valuation for B's house	Valuation for C's house
x	9	3	1
y	7	5	3
z	5	4	8

c.

Buyer	Valuation for A's house	Valuation for B's house	Valuation for C's house
x	1	8	4
y	7	4	3
z	6	3	4

5 Problem 5: Job Market

Problem. Suppose that there are a set of 3 employees in a job market, A-C, and a set of 3 employers, D-F. The valuations of the employers for the employees are as follows (x means unknown).

	Value for Employee A	Value for Employee B	Value for Employee C
Employer D	6	3	6
Employer E	7	2	x
Employer F	5	4	6

Also, suppose that Employee A is asking for a salary of 3, Employee B is asking for a salary of 1, and Employee C is asking for a salary of 5. You can think of the salaries as the prices in this market.

a. What is the smallest value of x such that the given set of salaries (3 for Employee A, 1 for Employee B, and 5 for Employee C) are market-clearing? Explain.

b. Under the value of x found in (a), what is the total payoff? What is the total valuation in the optimal assignment of employers to employees?